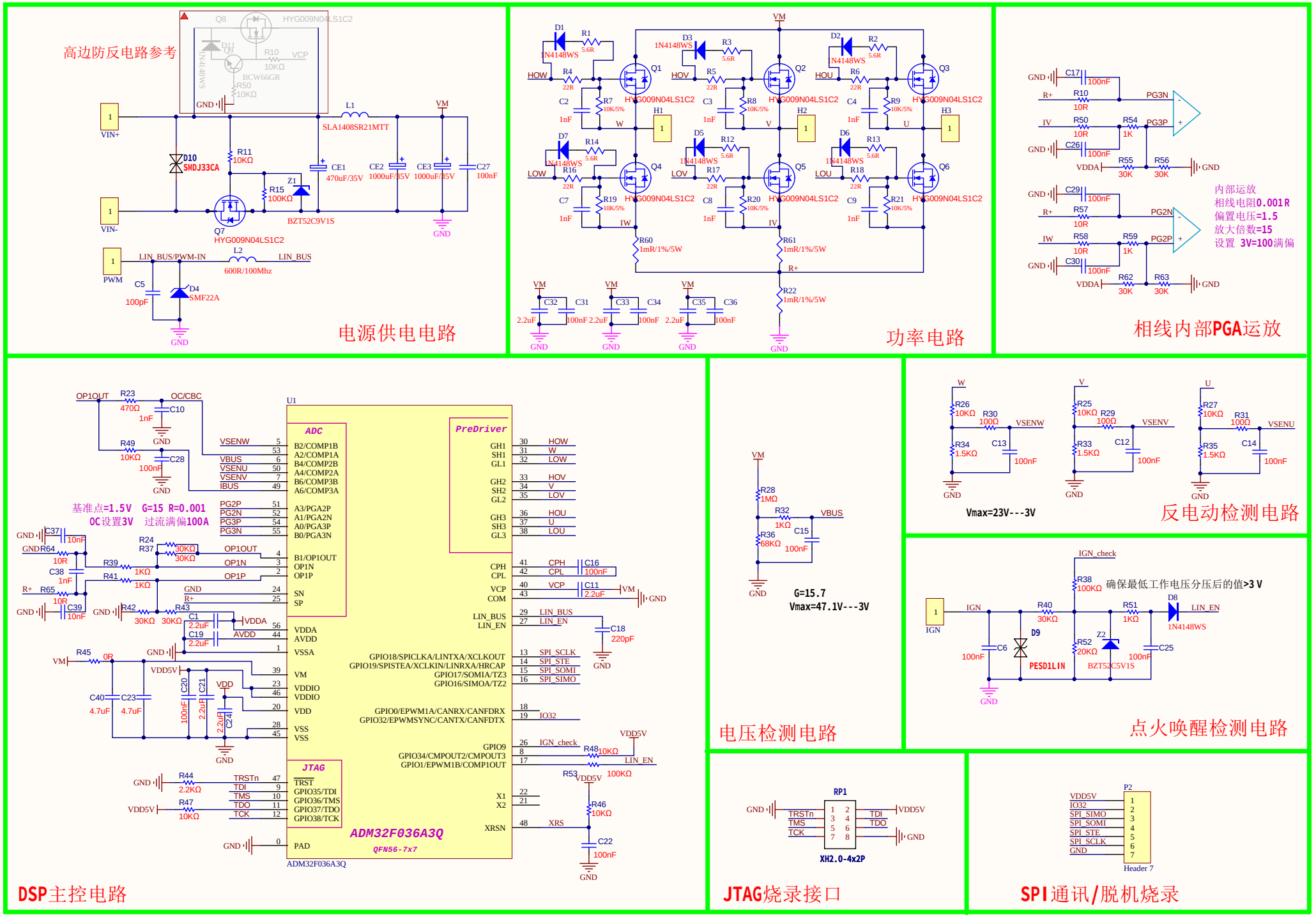

高边防反电路参考
VIN+
VIN-
PWM
D10
SMDJ33CA
R11
10KΩ
R15
100KΩ
Z1
BZT52C9V1S
CE1
470uF/35V
CE2
1000uF/35V
CE3
1000uF/35V
C27
100nF
L1
SLA1408SR21MTT
Q7
HYG009N04LS1C2
LIN_BUS/PWM-IN
LIN_BUS
600R/100Mhz
C5
100pF
D4
SMF22A
GND
VM
电源供电电路
D1
D2
D3
D4
D5
D6
1N4148WS
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
R15
R16
R17
R18
R19
R20
R21
5.6R
22R
10K/5%
HOW
HOV
HOU
LOW
LOV
LOU
Q1
Q2
Q3
Q4
Q5
Q6
HYG009N04LS1C2
H1
H2
H3
W
V
U
IW
IV
IU
C2
C3
C4
C7
C8
C9
1nF
R60
R61
R22
1mR/1%/5W
R+
C31
C32
C33
C34
C35
C36
2.2uF
100nF
功率电路
C17
C29
C30
R10
R50
R54
R55
R56
R57
R58
R59
R62
R63
10R
1K
30K
PG3N
PG3P
PG2N
PG2P
VDDA
内部运放
相线电阻0.001R
偏置电压=1.5
放大倍数=15
设置 3V=100满偏
相线内部PGA运放
U1
ADC
PreDriver
ADM32F036A3Q
QFN56-7x7
ADM32F036A3Q
B2/COMP1B
A2/COMP1A
B4/COMP2B
A4/COMP2A
B6/COMP3B
A6/COMP3A
A3/PGA2P
A1/PGA2N
A0/PGA3P
B0/PGA3N
B1/OP1OUT
OP1N
OP1P
SN
SP
VDDA
AVDD
VSSA
VM
VDDIO
VDD
VSS
PAD
GH1
SH1
GL1
GH2
SH2
GL2
GH3
SH3
GL3
CPH
CPL
VCP
COM
LIN_BUS
LIN_EN
GPIO18/SPICLKA/LINTXA/XCLKOUT
GPIO19/SPISTEA/XCLKIN/LINRXA/HRCAP
GPIO17/SOMIA/TZ3
GPIO16/SIMOA/TZ2
GPIO0/EPWM1A/CANRXA/CANFDRX
GPIO32/EPWMSYNC/CANTXA/CANFDTX
GPIO9
GPIO34/CMPOUT2/CMPOUT3
GPIO1/EPWM1B/COMP1OUT
X1
X2
XRSN
JTAG
TRST
GPIO35/TDI
GPIO36/TMS
GPIO37/TDO
GPIO38/TCK
IO32
OP1OUT
OC/CBC
VSENW
VBUS
VSENU
VSENV
IBUS
基准点=1.5V G=15 R=0.001
OC设置3V 过流满偏100A
PG2P
PG2N
PG3P
PG3N
SPI_SCLK
SPI_STE
SPI_SOMI
SPI_SIMO
IGN_check
TRSTn
TDI
TMS
TDO
TCK
DSP主控电路
R28
1MΩ
R32
1KΩ
R36
68KΩ
C15
100nF
VBUS
G=15.7
Vmax=47.1V---3V
电压检测电路
R26
R30
R34
R25
R29
R33
R27
R31
R35
10KΩ
100Ω
1.5KΩ
C13
C12
C14
100nF
VSENW
VSENV
VSENU
Vmax=23V---3V
反电动检测电路
IGN_check
IGN
R38
100KΩ
确保最低工作电压分压后的值>3 V
R40
30KΩ
R51
1KΩ
D8
1N4148WS
LIN_EN
D9
PESD1LIN
Z2
BZT52C5V1S
R52
20KΩ
C6
C25
100nF
点火唤醒检测电路
RP1
XH2.0-4x2P
JTAG烧录接口
P2
Header 7
SPI通讯/脱机烧录